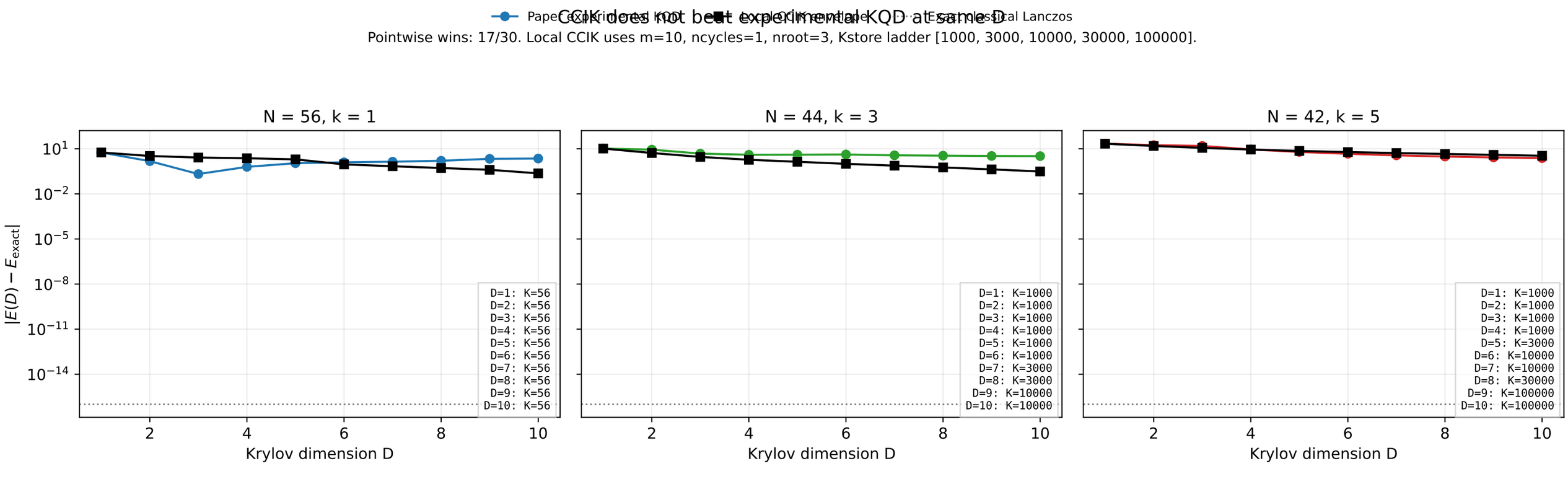
CCIK does not beat experimental KQD at same D
Paper CCIK
Original KQD
Experimental KQD
Exact diagonalization
Numerical Lanczos
Pointwise wins: 17/30. Local CCIK uses m=10, ncycles=1, nroot=3, Kstore ladder [1000, 3000, 10000, 30000, 100000].
N = 56, k = 1
N = 44, k = 3
N = 42, k = 5
$|E(D) - E_{exact}|$
10^{1}
10^{-2}
10^{-5}
10^{-8}
10^{-11}
10^{-14}
D=1: K=56
D=2: K=56
D=3: K=56
D=4: K=56
D=5: K=56
D=6: K=56
D=7: K=56
D=8: K=56
D=9: K=56
D=10: K=56
D=1: K=1000
D=2: K=1000
D=3: K=1000
D=4: K=1000
D=5: K=1000
D=6: K=1000
D=7: K=3000
D=8: K=3000
D=9: K=10000
D=10: K=10000
D=1: K=1000
D=2: K=1000
D=3: K=1000
D=4: K=1000
D=5: K=3000
D=6: K=10000
D=7: K=10000
D=8: K=30000
D=9: K=100000
D=10: K=100000
2
4
6
8
10
Krylov dimension D